Feedback on COMP40018a Exam (paper 1)

Question 1

This year’s exam has been of average difficulty with respect to previous years, but had certain parts that a lot of students found really difficult. The main point I would like to make is that many students showed poor time management. Very few students answered all the parts, thereby not giving the opportunity to gain marks. Please bear in mind that we cannot give any marks to an empty answer, and that spending a lot of time on one part cannot compensate for not answering another.

Question 1 - Average performance: B

1a) This part asked for some set construction, testing the understanding of the basic set operations; most students got this right.

1b) This was a variant of an exercise in the notes; most students got this right, but some forgot to check for symmetry after the transitive closure.

1c) This asked students to prove that the given relation is a partial order, and required understanding of what steps are allowed in proofs. Most students did reasonably well, but skipped steps in the proof; especially the part for reflexivity proved tricky. There were quite a few students who wrote an answer that was too sketchy, so did not constitute a proof, thereby dropping marks.

1d) This was an exercise on counting functions between two sets, a variant of an exercise in the notes, and most students did well.

1e) This part asked students to show a set was countable; a lot of students skipped this part, most who attempted it had difficulty writing a complete answer, attempting a listing diagram that was not well defined or correct.

Question 2

Question 2 - Average performance: B

2a.i) Most students answered this part of the question correctly. Remember the translation is into propositional logic, so no quantifiers nor variables! A few forgot to write the subformulas.

2a.ii) The aim of this part of the question was to ensure students were able to apply the equivalence laws correctly to rewrite the formula in the form requested. A few common mistakes included not writing the equivalence rule applied, or skipping a few steps. The formula $\neg(\phi 1 \wedge \ldots \wedge \phi n)$ is not in CNF nor is $\neg(\phi 1 \vee \ldots \vee \phi n)$.

2a.iii) The purpose of this part of the question was to test student's ability to construct proofs using direct argument clearly and concisely. The proof must be in both directions. Some students chose to prove that $\boldsymbol{\varphi} \leftrightarrow \boldsymbol{\psi}$ is a valid formula iff $(\boldsymbol{\varphi} \wedge \neg\boldsymbol{\psi}) \vee (\neg\boldsymbol{\varphi} \wedge \boldsymbol{\psi})$ is unsatisfiable. That approach should not be used unless you have shown $\vDash \boldsymbol{\varphi} \leftrightarrow \boldsymbol{\psi}$ is a valid argument (or it is explicitly stated that it could be used). Remember to justify every step in the proof and every conclusion derived. Common errors included forgetting to prove the other direction or missing justifications for steps in the proof (like referring to semantics of operators). Others are in which answers considered every possible valuation to $\boldsymbol{\varphi}$ and $\boldsymbol{\psi}$.

2b.i)(A) Most students answered this part of the question well. The main error was in not considering nodes that had no outgoing edges as possible values that would make a formula vacuously true.

2b.i)(B) The main error in this part of the question was in not differentiating the cases for which the relation F must have been true (no other possible truth value was possible) from when it could have been true (i.e. the false case was also a possibility).

2b.ii) This part of the question had the most wrong answers. Please make sure that when you open a ND box you know what is the assumption and what is the conclusion that you have to prove within that box before developing the proof inside the box. Most of the wrong answers were because the derivation lacked structure and students got lost along the proof.

Feedback on COMP40018 Exam (paper 2)

Question 1

This question's primary remit was to practise the various forms of proof by induction, and as secondary remit the precise development of such proofs, and finding stronger properties when reasoning about programs. The questions were demanding, and the class did well.

Question 1 - Average performance: C

1a) This part of the question had the aim to give an intuition how the functions **P1** and **P2** work, and when they terminate. Almost all students did this part perfectly

1b) The fist challenge was to determine that we would apply induction on m-n — most students saw that. The second challenge was to set up the structure of the proof (what is assumed, and what is to be shown), and in particular to see the need to for m and n auto be universally quantified both in what is to be proven in the base case, in the inductive hypothesis, and in what is to be shown in the inductive step. Some students did not do that correctly. The last challenge was developing the proofs correctly, i.e. say what is taken arbitrary, and justify each step.

1c) This follows from part b — a giveaway is the low number of marks for this part of the question. Most students saw that.

1d) The first challenge here was to see that in order to prove termination we have to apply induction on the function **P2** — not on its arguments. Most students saw that. The second challenge was to apply the induction on the definition correctly — some students made errors in the inductive step.

1e) The aim was to develop intuition as to how **split** and **zip** work. Almost all students did that correctly.

1f) The challenge here was to find the stronger lemma. That lemma would replace the empty character list in the result of **zip** by a non-empty character list (most students saw that), and then identify the relation between this nonempty resulting character list with the input list to **zip**, and the output list of **split** (many students had difficulty with that).

1g) This lemma can be proven by induction on the input tree (t) or on the output shape (st). The former is easier, while the latter is more complex as it also requires auxiliary lemmas. Many students set up the structure of the proof correctly, but some omitted necessary quantifiers (all input/output parameters except the variable on which we apply induction need to be universally quantified). The remaining challenge was developing the proofs correctly, i.e. say what is taken arbitrary, and justify each step.

Question 2

Great job overall. As a class you performed very well on this question, although some students clearly struggled to budget their time effectively between the two questions on the second exam paper.

Question 2 - Average performance: B

Here are some more details:

In part 2a) most students were able to evaluate the example method-calls correctly, although there were a few mistakes with the final **len** values returned.

In part 2b) almost all students were able to identify that there was an issue with empty array inputs and provide a correct update to the method's precondition to rule out such a case.

Part 2c) was answered very well on the whole, especially the mid-condition and variant parts. However, a lot of students struggled to correctly describe the unmodified part of the array or get exactly the right bounds on the array-slices used in the ***Abbreviates*** predicate.

Quite a lot of students skipped part 2d), or ran out of time for it. For those that did attempt the part, most were able to set-up the proof correctly, including the use of $_{old}$ annotations in the appropriate places. It was rare to see an attempt at the proof body, but those that did try tended to get at least half of the method marks for that aspect of the assessment criteria, making self-consistent proof steps, even if the proof set-up wasn't entirely correct. Showing that the **pos** and **cnt** variables stayed within bounds was achieved by most students attempting the proof body, but very few students were able to manipulate the array slices into the correct format to apply the ***AB+*** lemma in strict rigour.